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Effects of Chinese herbal medicines on growth performance, intestinal flora, immunity and serum metabolites of hybrid grouper (*Epinephelus fuscoguttatus*♀×*Epinephelus lanceolatu*♂)

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ABSTRACT

The immunomodulatory roles of Chinese herbal Medicine (CHM) in aquatic animals have been well-recorded. However, how CHM impacts the intestinal microbiota and serum metabolism is not fully understood. In this study, the effects of different additive levels of CHM on the growth performance, immunity, intestinal flora and serum metabolism of hybrid grouper (♀*Epinephelus fuscoguttatus* × ♂*Epinephelus lanceolatu*) were investigated. The addition of 0.5%, 1.0%, 1.5% and 2.0% Chinese herbal medicine compound to feed could significantly improve the weight gain rate (WGR), specific growth rate (SGR) and survival rate (SR) of grouper, reduced feed coefficient, while had no significant difference on morphometric parameter. The most significant improvement for the parameters above was observed in 1.5% group. Different addition levels of CHM could also significantly enhance the activities of ACP, AKP, SOD, CAT and LZM in serum. Accordingly, the supplementation of CHM significantly induced up-regulation of immune genes such as IL-8, IL-1β, TNF-α, Nrf2, Lzm in the liver, spleen and head kidney of grouper, improved the resistance of grouper to *V. harveyi* as well. The intestinal flora analysis showed that at the phylum level, the main dominant species of intestinal microorganisms were *Firmicutes*, *Proteobacteria*, *Bacteroidota*, *Actinobacteriota*, *Gemmatimonadota*, *Desulfobacterota*, *Fusobacteriota* and *Myxococcota*. At the genus level, the high abundance was *Lactobacillus*, *Streptococcus*, *Bacteroides*, *Escherichia*, *Romboutsia*, *Sphingomonas* and *Muribaculaceae*. The abundance of probiotics (such as *Lachnospiraceae*, *Lactobacillaceae*, *Streptococcaceae*, etc) in CHM-supplement groups were higher (highest in 1.5% group) compared with control group. Moreover, a total of 11 common differential metabolic pathways were screened by LC-MS metabolism analysis of serum, they were Neuroactive ligand-receptor interaction, Purine metabolism, Linoleic acid, Glycerophospholipid metabolism, Taurine and hypotaurine metabolism, Arginine and proline metabolism, ABC transporters, Aminoacyl-tRNA biosynthesis, Arachidonic acid metabolism, Drug metabolism-cytochrome P450, alpha-Linolenic acid metabolism. Also, three common differential metabolites (PI(20:4(5Z,8Z,11Z,14Z)/18:1(11Z)), PC(20:3(8Z,11Z,14Z)/22:1(13Z)), PC(22:0/20:4(5Z,8Z,11Z,14Z)) associated with intestinal health, growth and disease resistance was found. These data will contributes to a comprehensive understand for the regulatory roles of CHM on fish, which is also beneficial for the disease control and sustainable development of aquaculture.

1. Introduction

Chinese herbal medicine (CHM) has been widely used as feed additive in aquaculture because of its natural and cost-effective. Many studies have recorded that CHM contributes to grow and disease resistant of fish. Eakapol Wangkahart et al. found that kumquat fruit extracts

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could promote the growth of tilapia through enhancing digestive enzyme activity [1]. ED Abarike et al. showed that the addition of CHM significantly increased the weight gain and specific growth rate of tilapia, reduced feed coefficient, improved immune response and disease resistance of tilapia [2]. Huang et al. found that CHM could not only significantly improve the intestinal digestive enzyme activity and growth performance, but also significantly improved serum non-specific immunity of European eel [3]. Choi et al.'s research indicated that CHM protected grey mullet from *Lactococcus garvieae* infection through improving bactericidal activity and immunoglobulin level [4].

Except for the growth performance and immunity, emerging evidence have revealed the regulatory roles of CHM on intestinal flora of fish [5]. The research of Meng et al. showed that the addition of yam peel to the feed could enhance the immunity of carp by regulating the diversity and structure of intestinal flora [6]. Also, addition of cinnamic acid could increase the intestinal flora associated with sugar metabolism and amino acid metabolism, and promoted the growth of zebrafish [7]. However, the comprehensive roles of intestinal microbiota and metabolism after CHM supplementation on fish remain largely unclear.

Pearl gentian grouper (♀*Epinephelus fuscoguttatus* × ♂*Epinephelus lanceolatu*) is a new aquacultural marine fish in China. But the outbreaks of *vibriosis* caused by *Vibrio* spp have restricted the development of the industry [8]. Thus it is important to figure out the approach for prevention and control of *vibriosis.* To date, although the immunomodulatory roles of feed additive such as vitamin E [9], inositol [10], n-3 highly unsaturated fatty acids [11]on hybrid grouper have been recorded, the effects of CHM-supplement on hybrid grouper remain unknown. In this study, the different additive levels of CHM on the growth performance, immunity, intestinal flora and serum metabolism of the hybrid grouper (♀*Epinephelus fuscoguttatus* × ♂*Epinephelus lanceolatu*) were investigated. These data will provide a reference for the development of new strategies for *vibriosis* control in marine culture.

2. Materials and methods

2.1. Chinese herbal medicine additives and experimental feeds

Chinese herbal medicine compound is composed of astragalus, angelica, hawthorn, Licorice root and honeysuckle [12]. The materials were purchased from medicine market of Zhanjiang City, Guangdong Province. The materials were washed with double distilled water and dried in the air. After being crushed by a pulverizer, they were sieved with 80 meshes, mixed with proper proportion, and stored in a sealed bag at 4 °C. Before making the experimental feed, grind the basic feed, add it into the basic feed according to the amount (w/w) of 0.0% (control), 0.5%, 1.0%, 1.5% and 2.0%, fully mix it to make a 3 mm sinking feed, air dry it for 48h at room temperature, seal the bags for sub packaging and store it at −20 °C.

2.2. Experimental fish and management

The experimental fish was purchased from a fish farm on Donghai island, Zhanjiang city, Guangdong Province. Before experiment, the purchased groupers were temporary cultured in a pond (5.0 m × 3.0 m × 2.5 m) for one week and kept feeding with basic feed. Then the fish were fasted for 24h and the healthy grouper (22.0 ± 1.00g) were selected and randomly allocated to 0.3 m^3 breeding tanks with 30 fish in each tank, and 3 replicates were set for control and each experimental group (0.5%, 1.0%, 1.5% and 2.0% groups). During another 4 weeks breeding, the fish were fed according to 3% - 5% of their weight, and were fed with apparent satiety twice a day (8:30 and 17:30). During the breeding period, 50% water change was carried out every day.

2.3. Determination of growth performance

Serial parameters for growth performance were evaluated:

Table 1
Primers used in this study.

Primer	Sequences (5′-3′)
IL8-F	AGTCATTGTCATCTCCATTGCG
IL8-R	AAACTTCTTGGCCTGTCCTTTT
IL1β-F	CCTCATCATCGCCACACAGA
IL1β-R	TGCCTCACAACCGAACACAT
TNFα-F	TTCTCTTTGGCCTGATTGC
TNFα-R	GCTTCTTTTCCTGCGTCTGTTG
Nrf2-F	TATGGAGATGGGTCCTTTGGTG
Nrf2-R	GCTTCTTTTCCTGCGTCTGTTG
Lzm-F	ATGGGACAGTGAGGAACACC
Lzm-R	GTTGCGGATGCGTCCAATAA
β-actin-F	GGACTCTGAGCGCCGTC
β-actin-R	GTGCGGCGATTTCATCTTCC

Weight gain rate, WGR = 100 × (final average weight initial average weight);

Specific growth rate, SGR = 100 × (mean weight at the end of ln-mean weight at the beginning of ln)/days of feeding;

Feed conversion ratio, FCR = feed intake dry weight/(final weight- initial weight);

Survival rate, SR = 100 × number of fish tails at the end of the experiment/ number of fish tails at the beginning of the experiment;

Hepatosomatic index, HSI = 100 × liver weight/body weight;

Intestosomatic index, ISI = 100 × intestine weight/body weight;

Viserosomatic index, VSI = 100 × viscera weight / whole body weight

Condition factor, CF = 100 × body weight/body length.

2.4. Sample collection

2.4.1. Non-specific immune parameters and immune genes expression

The samples for immunity evaluation were collected on the 7th, 14th, 21st and 28th days, respectively. After fasting for 24 h, three fish from each group were randomly selected, then anesthetized and blood from each fish was drawn intravenously. For serum separation, the blood was clotted for 1h at room temperature and transferred to 4 °C for overnight. The blood clots were then centrifuged at 3000 rpm for 10 min. The supernatant was collected and stored at −80 °C for further analysis. The head kidney, liver and spleen tissues of each fish were sampled respectively and quickly frozen in liquid nitrogen and stored at −80 °C for determination of immune genes expressions.

2.4.2. Intestinal microbiota analysis

After 28 days for feeding, 6 fish were randomly selected from each experimental group, and their intestines were taken. After weighing, intestine of collected fish were quickly divided into RNA free tubes for quick freezing in liquid nitrogen, and stored at −80 °C for 16S sequencing. The microbial diversity in intestine was estimated using the alpha diversity that include Chao1 index and Shannon index. The unweighted Unifrac Principal coordinates analysis (PCoA) was performed by R package to estimate the beta diversity. Then the R package was used to analyze the significant differences between different groups using Wilcoxon statistical test.

2.5. Determination of non-specific immune parameters in serum

The activities of SOD, ACP, AKP, LZM, CAT in serum were assayed through corresponding commercial detection kits (Nanjing Jiancheng Bioengineering Institute, Nanjing, China) according to user's manual.

2.6. Detection of immune-related gene expression

Expression levels of immune-related genes including IL-8, IL-1β, TNF-α, Nrf2, Lzm were detected through real-time qPCR (RT-qPCR). β-actin was used as an internal reference gene.The used primers were shown in Table 1. The procedures of RNA extraction, cDNA synthesis, and RT-qPCR for analysis of immune gene expression were carried out as previously described (Wei et al., 2020).

2.7. *Vibrio harveyi* challenge

2.7.1. LD_{50} determination

Vibrio harveyi used for challenge was isolated and kept in our lab (Wei et al., 2020). The frozen bacteria was restored and cultured at 28 °C with 120 rpm/min. The proportion for activation was bacterial solution: medium = 1:100. Counting the number of bacteria and set 6 concentration gradients (10, 10^2, 10^3, 10^4 times, respectively). Total fifty fish were divided equally into one control group and four experimental group (10 fish/group). In the experimental group, 200 μL bacterial solution was injected intraperitoneally into each fish. The control group was injected with 200 μL PBS at the same manner. The number of dead fish per day in each group was recorded for 14 days after injection.

2.7.2. Challenge experiment

The challenge experiment was carried out after the feed containing compound Chinese herbal medicine was continuously fed for 28 days. The twenty fish in each group were selected for intraperitoneal injection of 200 μL (5.54×10^8 cfu/mL) bacterial solution. All fish were fed with basic feed after challenge and recorded the number of death fish per day in each group for 14 days. The relative percent survival (RPS) calculated as follows: RPS (%) = [1-(cumulative mortality of immunized group/ cumulative mortality of control group) × 100%.

2.8. Serum metabolic analysis

Global metabolic analysis was conducted by using ultrahigh performance liquid chromatography-mass spectrometry (UPLC-MS). UPLC-MS analysis was conducted with Waters Acquity UPLC I-Class plus UPLC and Thermo Fisher QE Ultra High Definition spectrometer. The QCs were injected at regular intervals (every 10 samples) throughout the analytical run to provide a set of data from which repeatability can be assessed.

The acquired LC-MS raw data were analyzed by the progqenesis QI software (Waters Corporation Milford, USA) using the following parameters: precusor tolerance was set 5 ppm, fragment tolerance was set 10 ppm, and retention time (tolerance was set 0 02 min Internal standard detection parameters were deselected for peak RT alignment, isotopic peaks were excluded for analysis, and noise elimination level was set at 10.00, minimum intensity was set to 15 of base peak intensity. The differential metabolites were selected on the basis of the combination of a statistically significant threshold of variable influence on projection (values obtained from the OPLS DA model and p values from a two tailed Student's t-test on the normalized peak areas, where metabolites with VIP values > 1.0 and p values < 0.05 were considered as differential metabolites.

2.9. Statistical analysis

Statistical analyses were performed using SPSS 17.0 (SPSS Inc., USA). Group differences were determined by Duncan's test. Spearman rank correlation coefficients of different species and metabolites pairs were calculated using data from the same samples. ** indicates highly significant difference compared with the control group ($p < 0.01$). * indicates significant difference compared with the control group ($p < 0.05$).

Table 2
Effects of CHM-supplement on growth performance of grouper.

Group	IW (g)	FW(g)	WGR (%)	SR (%)	FCR	SGR (%)
Control	21.98 ± 0.04	39.24 ± 0.26^a	78.49 ± 1.15^a	86.67 ± 0.00^a	1.06 ± 0.01^a	2.07 ± 0.02^d
0.5%	21.96 ± 0.03	43.80 ± 2.90^b	99.46 ± 5.92^b	90.83 ± 3.54^b	0.84 ± 0.12^b	2.46 ± 0.19^c
1.0%	22.01 ± 0.00	44.78 ± 0.56^b	103.49 ± 2.51^b	93.33 ± 0.00^b	0.83 ± 0.02^b	2.54 ± 0.04^{bc}
1.5%	22.04 ± 0.04	49.24 ± 0.58^c	123.43 ± 3.08^c	95.00 ± 02.36^b	0.67 ± 0.02^c	2.87 ± 0.03^a
2.0%	22.01 ± 0.07	46.70 ± 0.56^{bc}	112.14 ± 1.89^{bc}	94.43 ± 0.04^b	0.76 ± 0.01^{bc}	2.69 ± 0.04^b

Table 3
Effects of CHM-supplement on morphometric parameters of grouper.

Group	VSI (%)	HSI (%)	CF (%)	ISI (%)
Control	10.56 ± 0.34	3.45 ± 0.87	2.58 ± 0.74	1.10 ± 0.14
0.5%	10.75 ± 0.30	3.72 ± 0.27	2.56 ± 0.73	1.02 ± 0.17
1.0%	10.41 ± 0.22	3.33 ± 0.22	2.53 ± 0.59	1.14 ± 0.11
1.5%	10.46 ± 0.30	3.42 ± 0.29	2.49 ± 0.73	1.05 ± 0.10
2.0%	10.71 ± 0.93	3.48 ± 0.34	2.61 ± 0.83	1.05 ± 0.19

3. Results

3.1. Effect of CHM on the growth performance of hybrid grouper

As shown in Table 2, at the 28 day of feeding, compared with the control group, the WGR, FW, SGR and SR of grouper fed with CHM-supplement were significantly increased, and the highest level was seen ($p < 0.05$) in 1.5% group, while the FCR in all CHM-supplement group were reduced. WGR in 0.5% group and 1.0% group was significantly higher than that in the control group, and significantly lower than that in 1.5% group. There was no significant difference between 1.5% group and 2.0% group. The survival rate of the experimental group was significantly higher than control group, and there was no difference between the experimental groups ($p < 0.05$). FCR in 0.5% group and 1.0% group was significantly higher than control group, and significantly lower than 1.5% group. There was no significant difference between 1.5% group and 2.0% group ($p < 0.05$). The SGR of all experimental groups were significantly higher than control group, and the SGR in 1.5% group was significantly higher than that of the other groups ($p < 0.05$). Additionally, there was no significant difference in VSI, HSI and ISI in CHM-supplement groups compared with control group (Table 3) ($p < 0.05$).

3.2. Immune-modulatory effects of CHM on hybrid grouper

3.2.1. Non-specific immune parameters in serum

The activities of non-specific immune parameters including AKP, ACP, LZM, SOD and CAT were determined. All detected parameters were induced after in CHM-supplement groups and the highest levels of were observed in 1.5% group at different time points (AKP and CAT at 21st day, ACP and LZM at 28th day) (Fig. 2).

3.2.2. Immune-related gene expressions in liver, spleen and head kidney

In order to evaluate the effects of CHM on immunity, a couple of immune-related genes (IL-8, IL-1β, TNF-α, Nrf2, Lzm) were chosen for detection through RT-qPCR. As shown in Fig. 1, compared with the control group, the expression levels of all examined genes were stimulated in liver, spleen and head kidney. The highest expression levels of IL-8 and Lzm were seen in 0.5% group, and the highest transcriptional levels of IL-1β, TNF-α and Nrf2 were observed in 1.0% group.

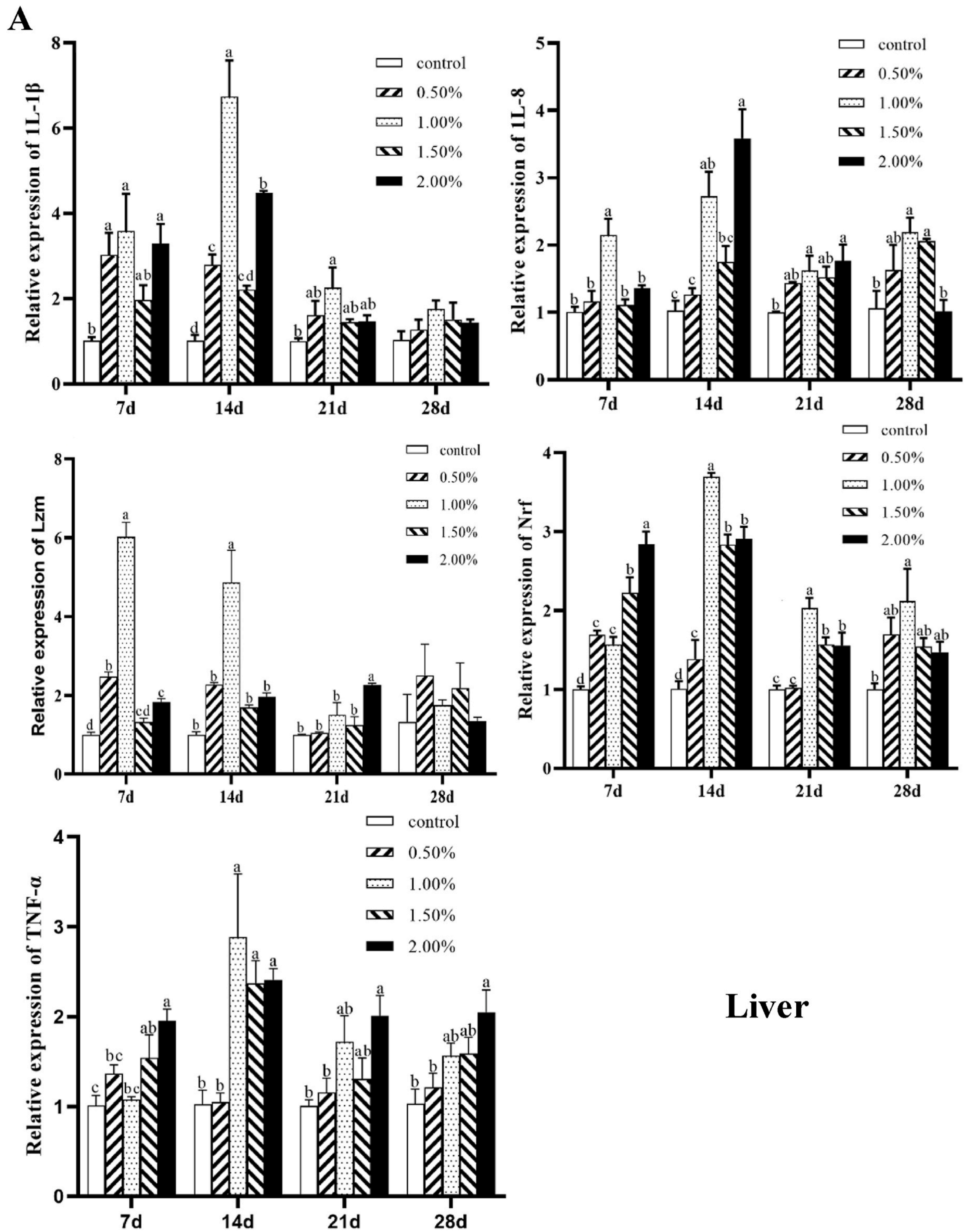

Fig. 1. Effects of CHM-supplement on the expression of immune genes in different immune organs (A: Liver, B: Spleen, C: Head kidney) of grouper. Different letters indicate significant differences.

3.2.3. Bacterial resistance of hybrid grouper

To determine the effects of CHM on bacterial resistance of hybrid grouper, the challenge experiment with *Vibrio harveyi* was carried out. The results exhibited that the relative protection rate (RPS) of 0.5% group, 1.0% group, 1.5% group, and 2.0% group respectively was significant higher than control group, of which the highest RPS was seen in 1.5% group (64.29%) (Table 4).

3.3. Effects of CHM on the intestinal flora of hybrid grouper

After feeding CHM, at the phyla level, relative abundance of top ten species were *Firmicutes*, *Proteobacteria*, *Bacteroidota*, *Actinobaciota*, *Gemmatimonadota*, *Desulfobacterota*, *Fusobacteriota*, *Myxococota*, etc. The relative abundance of each group is shown in Fig. 3 and Table 5. In the CHM-supplement group, the abundance of *Trichospiriceae*,

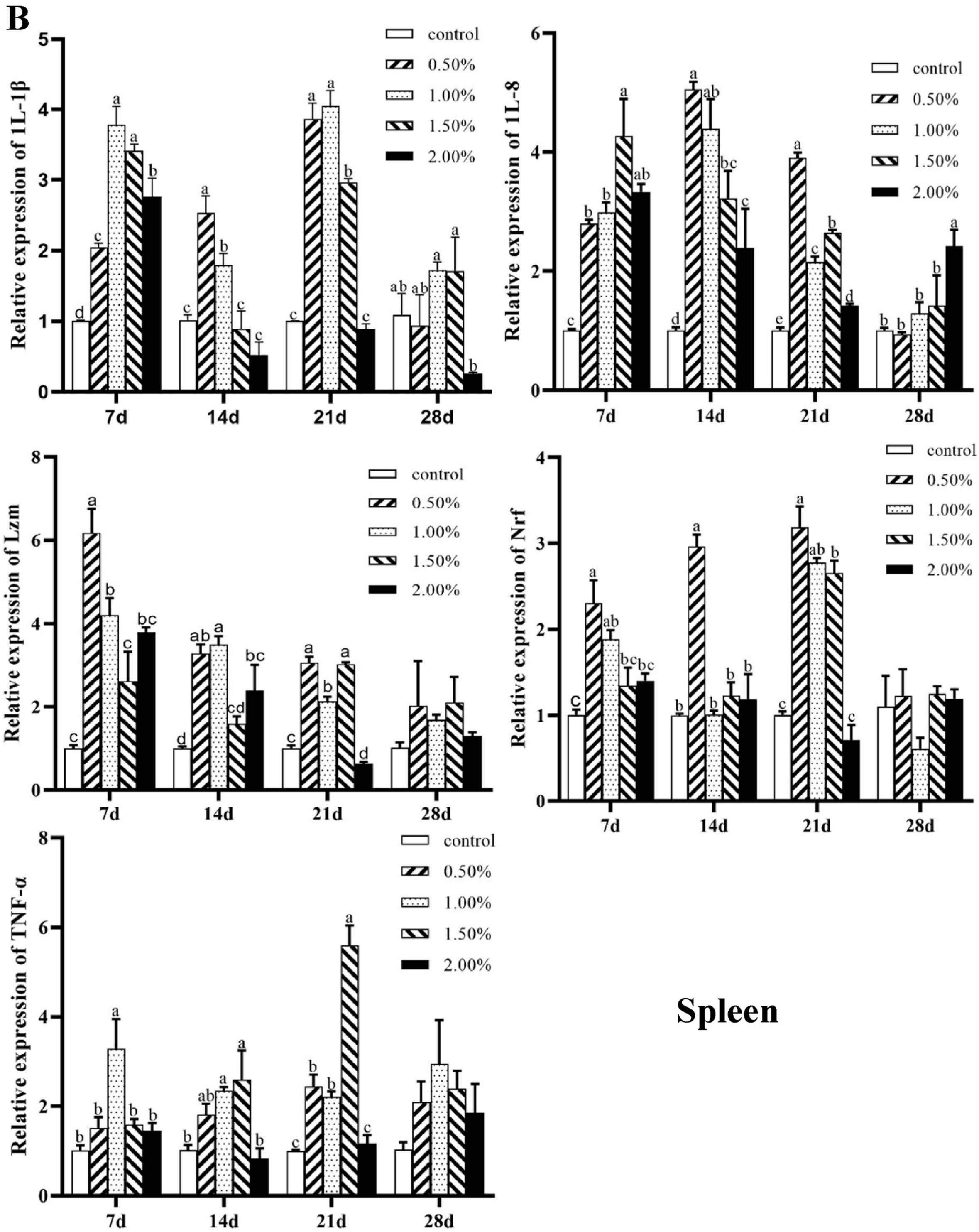

Fig. 1. (*continued*).

Lactobacilliceae and *Streptococcaceae* increased and the highest abundance was observed in 1.5% group ($p < 0.05$). The abundance of *Peptostreptococcaceae* and *Ruminococcaceae* was the highest in the 1.5% group, and the 2.0% group was lower than the control group ($p < 0.05$). Bur the abundance of *Enterobacteriaceae*, *Sphingomonas* and *Staphylococcaceae* decreased in all experimental groups.

3.4. Effects of CHM on serum metabolic profiling

As shown in Table 6, the quantity of differential metabolites between the CHM-supplement groups and the control group exhibited a increasing trend with the increasing addition level. No strong outliers were detected according to the PCA-X score plotted and a better distinction was made after plotting the OPLS-DA score without two outliers (Fig. 4 and Fig. 5). In 0.5% group, 37 differential metabolites were found, 4 differential metabolites were down-regulated and 33 were up-regulated; 1.0% group had 68 differential metabolites, 16 were down-regulated and 52 were up-regulated; in 1.5% group, 28 metabolites were up-regulated and 17 metabolites were down-regulated; in 2.0% group, 46 metabolites were up-regulated and 24 metabolites were

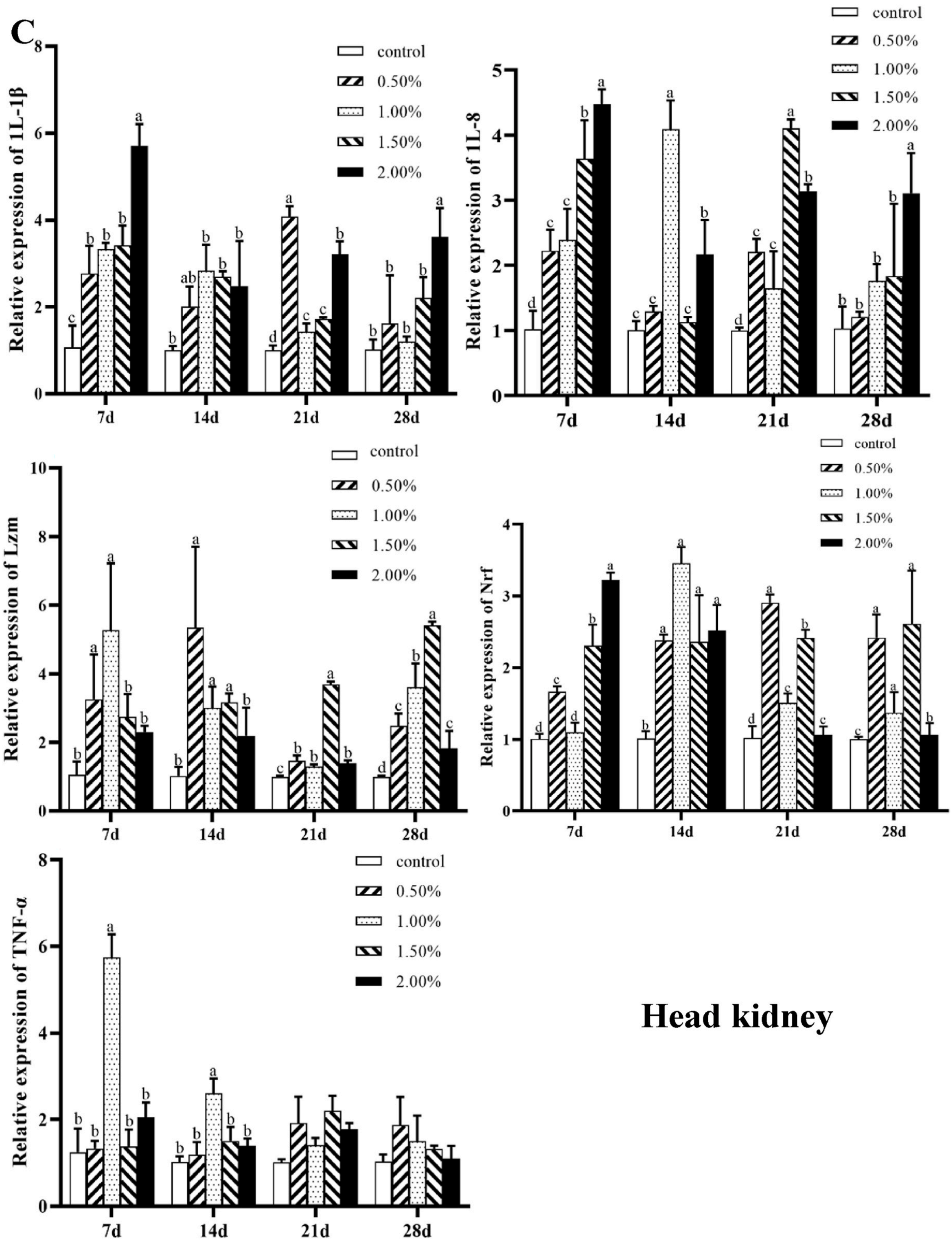

Fig. 1. (*continued*).

down-regulated. The 1.0%, 1.5%, 2.0% groups can be screened to phosphorylcholine PC (22:0/20:4 (5Z,8Z,11Z,14Z)), 12,13-DHOME, Isoleucyl proline (Isoleucyl proline), citric acid (Citric acid), myristic acid (Myristic acid) and other five metabolites. There were common differential metabolites among the CHM-supplement groups, they were phosphatidylinol PI(20:4(5Z,8Z,11Z,14Z)/18:1(11Z)), PC(20:3 (8Z,11Z,14Z)/22:1(13Z)), phosphocholine PC (22:0/20:4 (5Z,8Z,11Z,14Z)), all three differential metabolites were significantly up-regulated, and phosphocholine PC(22:0/20:4(5Z,8Z,11Z,14Z)) and the highest level were seen at the 2.0% group. Further KEGG pathway analysis showed that a total of 11 common differential metabolic pathways were involved: neuroactive ligands, purine metabolism, linoleic acid metabolism, glycerophospholipid metabolism, taurine metabolism, arginine and proline metabolism, ABC transport system, aminoacyl tRNA, arachidonic acid metabolism, drug metabolism-cytochrome P450, linolenic acid metabolism (Fig. 6 and Table 7).

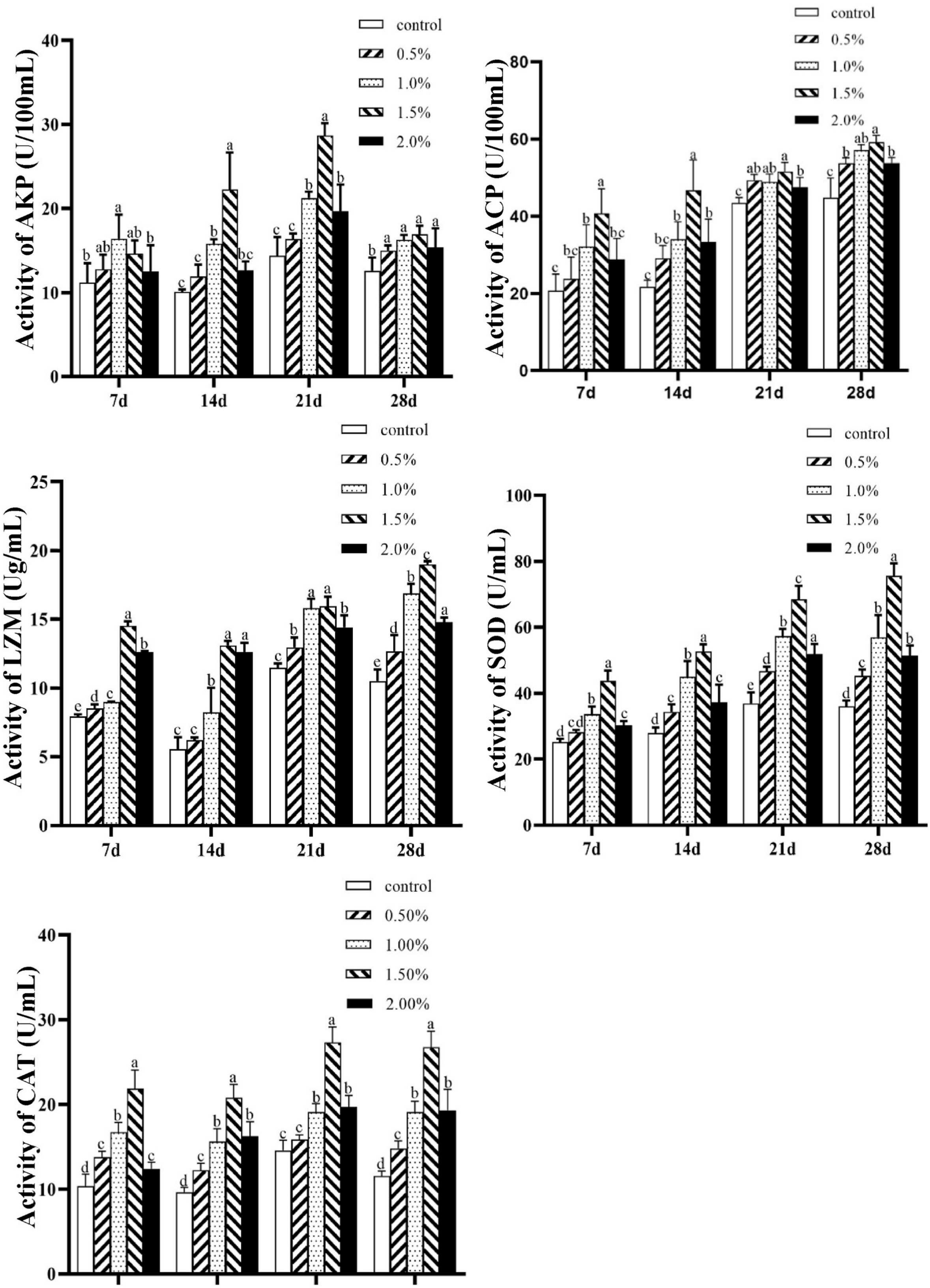

Fig. 2. Effects of CHM-supplement on the activity of serum non-specific immune parameters in grouper. Different letters indicate significant differences ($p < 0.05$).

Table 4
Relative percent survival in control group and CHM-supplement groups.

Group	Mortality rate	RPS
Control	70%	0
0.5%	40%	42.86
1.0%	30%	57.14
1.5%	25%	64.29
2.0%	45%	35.71

3.5. Correlation analysis between gut microbiota and serum metabolite

To clarify the interaction between gut microbiota and serum metabolites (the above 3 common metabolites), a correlation analysis was further performed (Fig. 7). PI(20:4(5Z,8Z,11Z,14Z)/18:1(11Z)) showed a negative correlation with Coriobacteriaceae_UCG-002 in 0.5 and 2.0 group, with Saccharopolyspora in 1.5 group. PC(20:3(8Z,11Z,14Z)/22:1 (13Z)) was negatively correlated with Saccharopolyspora in 0.5, 1.5 and 2.0 group, also with Coriobacteriaceae_UCG-002 0.5 and 2.0 group. Similarly, PC(22:0/20:4(5Z,8Z,11Z,14Z)) exhibited a negative correlation with Coriobacteriaceae_UCG-002 and Saccharopolyspora in both 1.0 and 2.0 group.

4. Discussion

It has been well-recorded that CHM can effectively promote the development of the digestive tract, accelerate nutrient absorption, increase the activity of digestive enzymes, and impact the intestinal flora of fish [13–15]. Similar with the previous reports, the current data showed that the supplement of the CHM (consisting of astragalus, hawthorn, angelica and isatis root) at different doses of 0.5%, 1.0%, 1.5%, and 2.0% could significantly increase the weight gain rate, specific growth rate and survival rate of the hybrid grouper, and reduce the feed coefficient. Like Ma et al. found that the addition of 1.88% *Andrographis paniculata* to feed could markedly increase the weight gain rate and specific growth rate of channel catfish [16]. He et al. fed the hybrid grouper with 40 g/kg supplement of *verbena* had the most significant effect on promoting growth, and 60 g/kg supplement of *wintergreen* had the best effect on improving immunity [17]. Adding 3‰ *Astragalus membranaceus* and *Phellodendron Phellodendri* could significantly improve the activity of lipase, the weight gain rate and specific growth rate of *loach Paramisgurnus dabryanus* [18]. In our experiment, the weight gain rate, specific growth rate and survival rate of the 1.5% group were significantly higher than the other groups, the feed coefficient was significantly reduced and should be the optimum dosage for the hybrid grouper culture.

To understand how CHM impacts the immune response of grouper (♀*Epinephelus fuscoguttatus* × ♂*Epinephelus lanceolatu*), the mRNA expression levels of immune-related genes, the activities of non-specific immune indexes and the resistance against *V.harveyi* were analyzed. After CHM supplementation, the immune genes including IL-8, IL-1β, TNF-α, Nrf2, Lysozyme (Lzm) were significantly up-regulated in the liver, spleen and head kidneys of grouper. IL-8 is a chemotactic cytokine that plays important role in immune response through modulating the activation of T lymphocyte and neutrophils [19]. IL-1β can activate macrophages and mediate in the interaction between leukocytes and other cell [20]. TNF-α regulates inflammatory response during the early stage of pathogen infection, enhances migration and killing activities of immune cells as well [21]. Nrf2 signaling pathway is the conserve anti-oxidant pathway that participated in fish immuno-nutritional

Table 5
Relative abundance of intestinal flora in control group and CHM-supplement groups.

Classification (Phylum level)	Control	0.5%	1.0%	1.5%	2.0%
Firmicutes	0.558 ± 0.01^{a}	0.554 ± 0.01a^{b}	0.538 ± 0.01a^{b}	0.586 ± 0.01b	0.568 ± 0.01^{ab}
Proteobacteria	0.2012 ± 0.001^{a}	0.1986 ± 0.016^{a}	0.1838 ± 0.00^{ab}	0.1573 ± 0.010^{c}	0.1440 ± 0.024^{C}
Bacteroidota	0.0942 ± 0.04	0.1102 ± 0.06	0.1267 ± 0.017	0.1177 ± 0.011	0.1137 ± 0.08
Actinobacteriota	0.1155 ± 0.04^{a}	0.1096 ± 0.04^{ab}	0.1160 ± 0.07^{ab}	0.1284 ± 0.011^{ab}	0.1371 ± 0.09^{b}
Gemmatimonadota	0.0188 ± 0.0016	0.0165 ± 0.0013	0.0167 ± 0.0022	0.0120 ± 0.0009	0.0166 ± 0.004

Table 6
Quantity of serum metabolites between different CHM-supplement groups compare with control group.

Group	Quantities of different serum metabolites
0.5%	37
1.0%	68
1.5%	45
2.0%	68

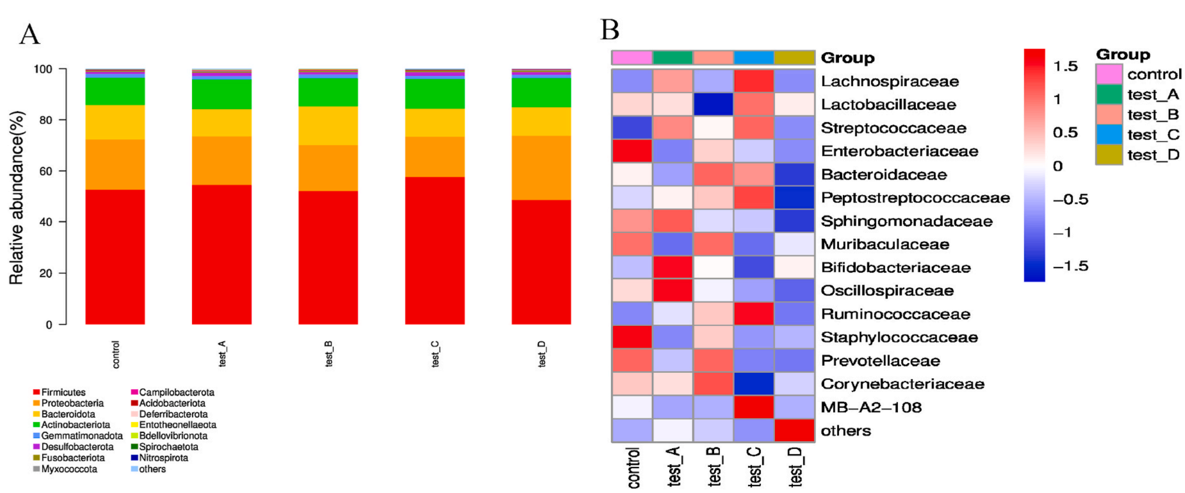

Fig. 3. Effect of CHM-supplement on intestinal flora of grouper. Comparison of the structure and abundance of intestinal flora of grouper after feeding TCM (A: phylum level, B: family level)
Note:test A-D represent addition TCM levels at 0.5% group, 1.0% group, 1.5% group, 2.0% group, respectively.

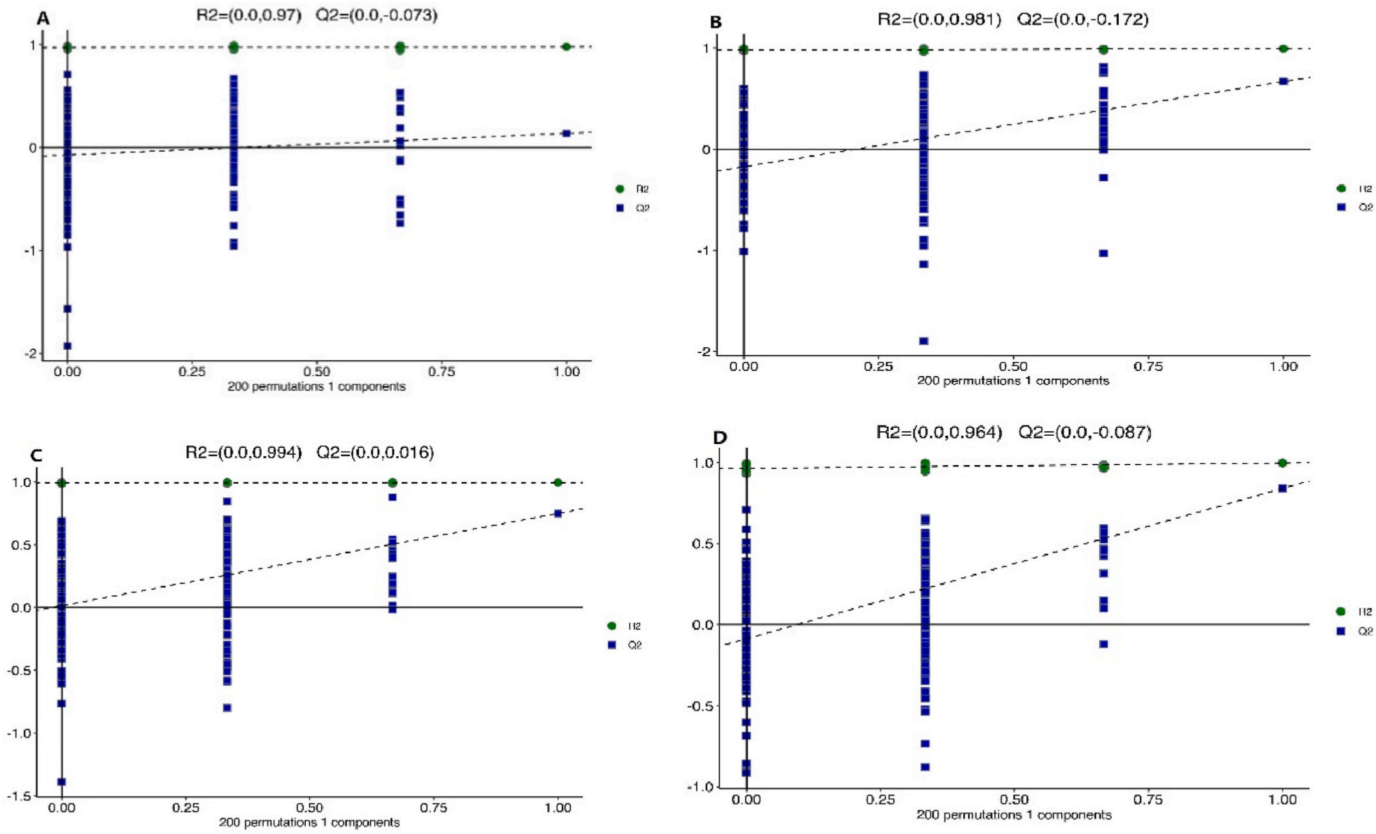

Fig. 4. PCA analysis of serum metabolite profiles CHM-supplement groups compare with control group. Note: A: 0.5% group, B: 1.0% group, C: 1.5% group, D: 2.0% group.

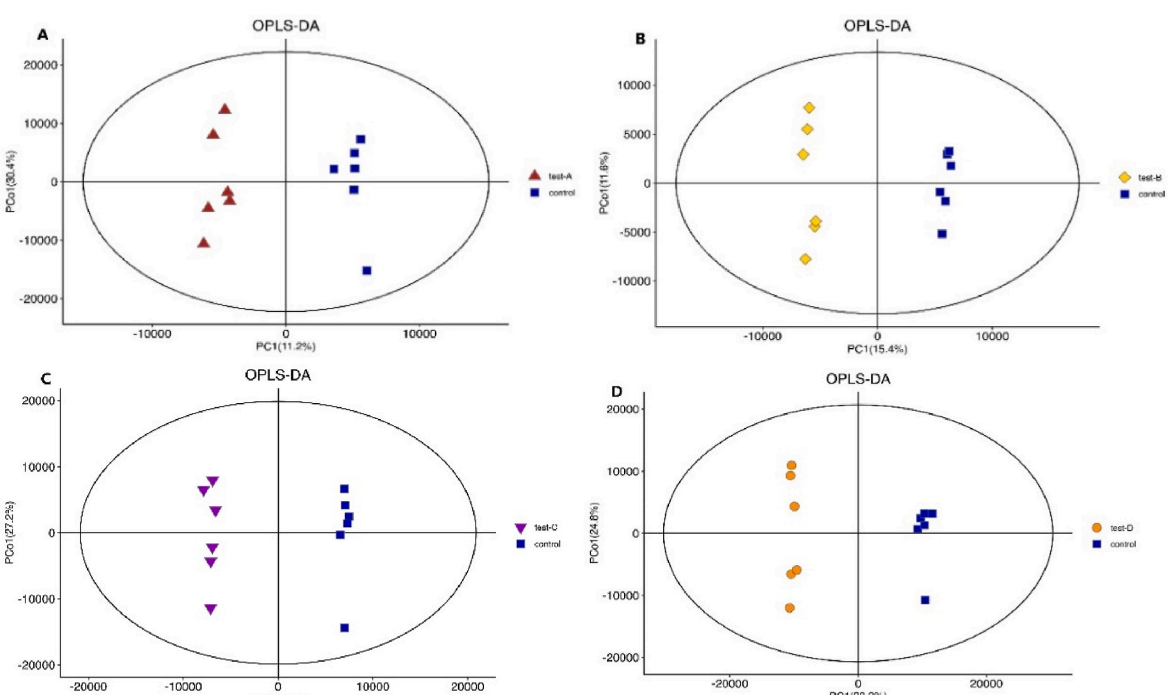

Fig. 5. OPLS-DA analysis of serum metabolite profiles of CHM-supplement groups compare with control group. Note: A: 0.5% group, B: 1.0% group, C: 1.5% group, D: 2.0% group.

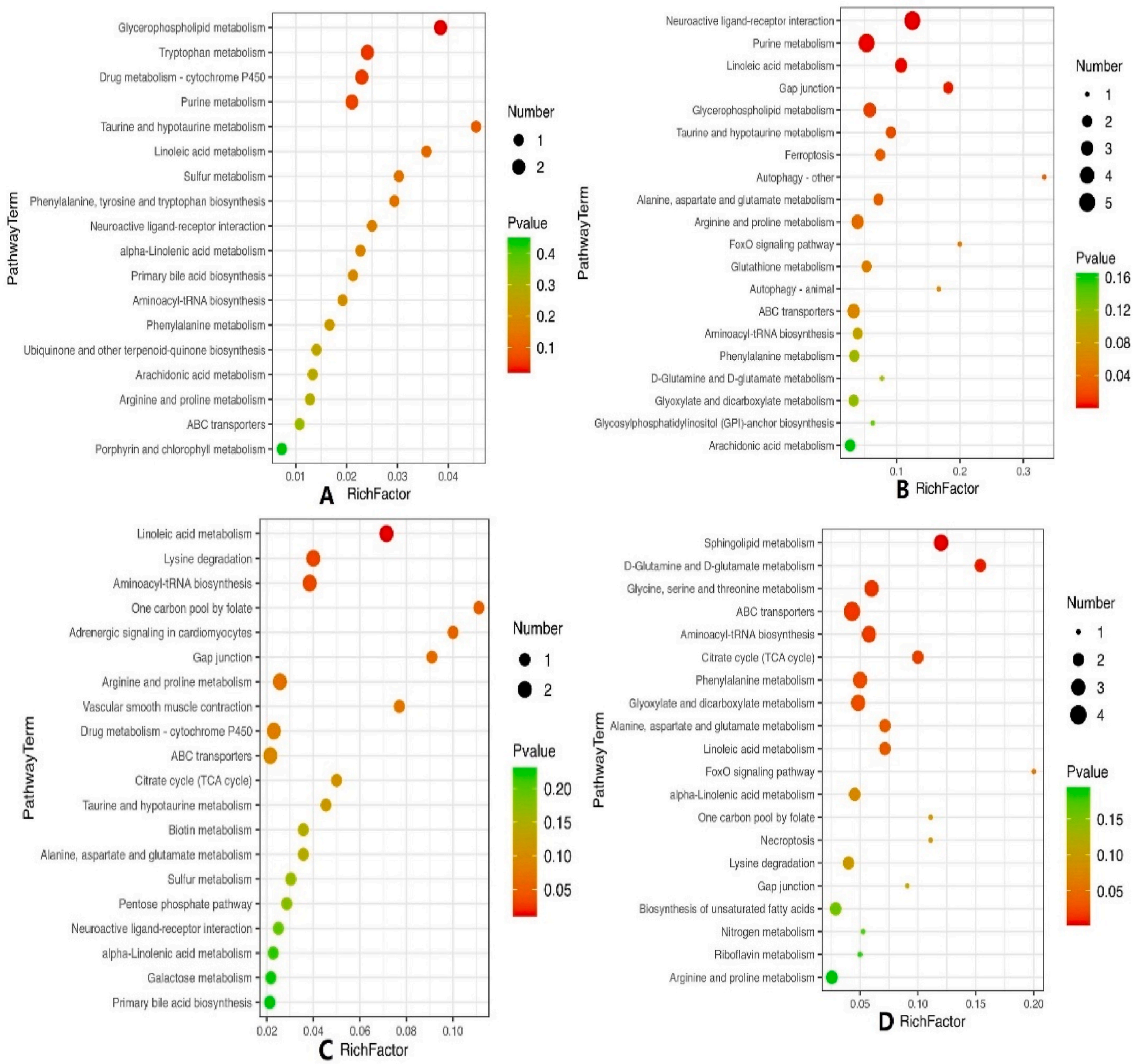

Fig. 6. The top 20 Metabolic pathway of differential metabolites from CHM-supplement groups compare with control group. Note: A: 0.5% group, B: 1.0% group, C: 1.5% group, D: 2.0% group.

Table 7
The common metabolic pathways in different CHM-supplement groups.

Number	Metabolic pathway
1	Neuroactive ligand-receptor interaction
2	Purine metabolism
3	Linoleic acid metabolism
4	Glycerophospholipid metabolism
5	Taurine and hypotaurine metabolism
6	Arginine and proline metabolism
7	ABC transporters
8	Aminoacyl-tRNA biosynthesis
9	Arachidonic acid metabolism
10	Drug metabolism - cytochrome P450
11	Alpha-Linolenic acid metabolism

response [22]. Lysozyme widely exists in aquatic animals and has well bacteriostasis activity against a variety of pathogenic bacteria [23]. Besides, CHM also improved several non-specific immune parameters including ACP, AKP, SOD, CAT and LZM of serum in this experiment. ACP and AKP can promote phagocytosis and intracellular digestion of phagocytes. SOD and CAT maintains cellular homeostasis by removing superoxide radicals [24–26]. These results suggested that CHM could improve the immunity and anti-oxidant responses of grouper, which were consistent with previous reports [2,18]. Furthermore, CHM-supplement supplied the grouper resistance to *V.harveyi* infection, the highest RPS (64.29%) was recorded in 1.5% group. The comprehensive evaluation above indicated that the CHM in this study could be applied as immune modulator for the control of fish *vibriosis* in aquaculture.

The former study has indicated that CHM can improve the growth

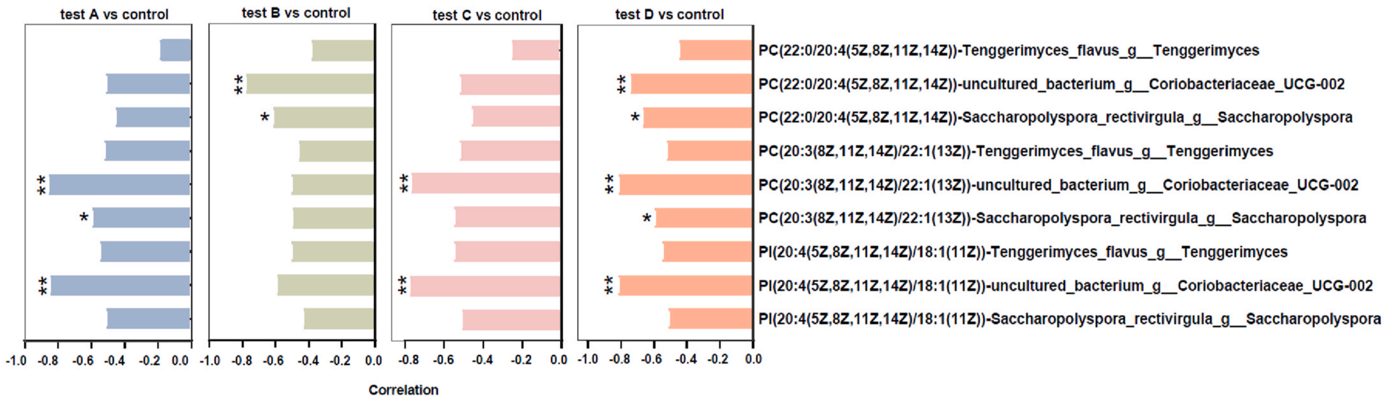

Fig. 7. The correlation analysis of differential metabolites from CHM-supplement groups compare with control group. Note: A: 0.5% group, B: 1.0% group, C: 1.5% group, D: 2.0% group.

and immunity by regulating the structure and diversity of intestinal flora [27]. The *Proteobacteria*, *Firmicutes*, *Fusobacteria* are the main groups in marine fish [28]. In this study, the similar results was observed and *Firmicutes* was accounted for the largest proportion at the phylum level. Additionally, the abundance of *Firmicutes* in 1.5% and 2.0% group were significantly higher than control group. *Proteobacteria* at 1.0%, 1.5% and 2.0% group were significantly lower than the control group. *Proteobacteria* conclude many pathogenic bacteria and it can be used as a indicator of intestinal flora imbalance [29]. The CHM used in this study has an inhibitory effect on the growth of proteobacteria so that maintain intestinal flora balance, which is beneficial to the health of fish. *Actinomycetes*, known as "treasure house of secondary metabolites", are a class of gram-positive bacteria that widely exist in natural environments and organisms that have good antibacterial activity [30,31]. In our study, the abundance of *Actinomycetes* at 2.0% group was significantly higher than the other groups, indicating CHM impacted intestinal flora by increasing the relative abundance of *Actinomycetes*. At the family level, the abundance of *Lachnospiraceae*, *Lactobacillusceae* and *Streptococcusceae* in the supplemental groups also increased and the highest level was seen in 1.5% group. *Lachnospiraceae*, *Lactobacillusceae* and *Streptococcusaceae* are important probiotic families in the body, which play an important role in the body's health and digestion [32]. These results indicated that CHM could alter the health and growth of grouper through modulating intestinal microbiota.

Based on LC-MS non-targeted metabolic analysis, three common serum metabolites including phosphatidylalcohol PI (20:4 (5Z,8Z,11Z,14Z)/18:1(11Z)), Phosphocholine PC (20:3(8Z,11Z,14Z)/22:1(13Z)) and Phosphocholine PC (22:0/20:4(5Z,8Z,11Z,14Z)) were screened in CHM-supplement groups, suggesting the three metabolites may be the potential effectors. Among them, phosphocholine PC (20:3 (8Z,11Z,14Z)/22:1(13Z)) and PC (22:0/20:4(5Z,8Z,11Z,14Z)) are one of the important components of cell biofilm [33]. Phosphocholine is one of the metabolites in choline metabolism, which plays a crucial role in cell proliferation and apoptosis [34]. The phosphorylcholine can also promote the synthesis of glutathione and enhance the antioxidant capacity [35]. Phosphatidylinositol is one of the metabolites of phosphatidic acid and participate in calcineurin activation, and T cell activation signal transduction [36]. The three common differential metabolites screened in current study are all involved in phospholipid metabolism, and an appropriate amount of phospholipids can improve the growth performance of rainbow trout [37]. Similarly, the addition of phospholipids to feeds can also stimulate growth and reduce feed coefficient and improve intestinal development of cobia [38] and tilapia [39]. Strikingly, the negative correlations of the metabolites above with Saccharopolyspora rectivirgula and Coriobacteriaceae UCG-002 were observed in different CHM-supplement groups. And it has been recorded that Saccharopolyspora rectivirgula induces lung inflammation [40]and Coriobacteriaceae UCG-002 is closely related to inflammatory bowel disease [41]. In addition, five common metabolites including 12, 13-DHOME, Isoleucyl proline, Citric acid, and Myristic acid were screened in all CHM-supplement groups. The 12,13-DHOME can be produced by intestinal bacterial cyclooxygenase, which can inhibit the secretion of anti-inflammatory factors and promote the inflammatory response [42]. Citric acid can improve digestive enzyme activity, nutrient absorption, protein utilization, intestinal flora and intestinal health, which has been widely used in aquaculture because of its enhanced roles on fish growth and immunity [43–45]. Among the 11 screened metabolic pathways in this study, the arachidonic acid metabolic pathway and related metabolites play an important role in immune regulation [46]. Taurine has various regulatory roles on antioxidant capacity, inflammatory cytokines expressions, lymphocyte differentiation and proliferation, etc [47,48]. The obtained metabolism data in this research indicated that CHM could improve intestinal health, growth and immunity of grouper through modulating serum metabolites.

In summary, the Chinese herbal Medicine (CHM) supplemented in this study is beneficial to the growth and immunity of hybrid grouper. The growth performance, activity of non-specific immune parameters, immune genes expression and resistance to bacterial pathogen of grouper are significantly improved after CHM supplementation, by regulating the intestinal flora and serum metabolic profiles. Thus the CHM developed in this study can be applied as a feeding additive for promoting the health and sustainable development of mariculture.

Declaration and verification

This manuscript to be considered for publication has not been published and is not under consideration for publication elsewhere.

CRediT authorship contribution statement

Jia Cai: Conceptualization, Formal analysis, Data curation, Writing – original draft, Writing – review & editing. **Zhenggao Yang:** Methodology, Project administration, Data curation. **Yu Huang:** Conceptualization, Formal analysis, Data curation. **Jichang Jian:** Investigation, Methodology, Resources. **Jufen Tang:** Investigation, Methodology, Resources.

Declaration of competing interest

The authors declare no competing interests.

Data availability

No data was used for the research described in the article.

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